# Photonic Devices - FBG report

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## I. Introduction

In this experiment, a measure of elongation and temperature shift is deduced from the reflected wavelength of a Fiber Bragg Grating reflector. The principle of operation of this device relies on the variation of the reflection response, and so the wavelength of maximum reflection, with elongation and temperature. The grating operation is based on the Fabry-Perot interferometer principle, from which, given a periodic perturbation pattern of the effective refractive index $n_{eff}$ of the propagation medium, of period $\Lambda$, we have a reflected spectrum peaked at the so called Bragg wavelength:

$$\lambda_B = 2n_{eff}\Lambda. \tag{1}$$

Given this peak wavelength, any variation in the period, or in the effective refractive index, will be measurable observing the reflected wavelength. The first quantity we are interested in is the strain, which is defined as the relative variation of the length of the fiber. The sensor we are considering is a lumped sensor, so it is only sensitive to the *relative* elongation of the fiber. The absolute elongation is of course easily determined once the fiber length is known. If the total length is $L$, the strain is defined as

$$\varepsilon = \frac{\Delta L}{L}. \tag{2}$$

The second quantity to be measured is the temperature: we focus on the measurement of the temperature difference from the rest condition $\Delta T$.

A simple model for the measure of strain and temperature of the environment in which the fiber is embedded, computed as a difference with respect to an unperturbed condition, is a first order approximation of the peak wavelength from the equation (1):

$$\Delta\lambda = \lambda_B\Big(\underbrace{\frac{1}{n}\frac{\partial n}{\partial \varepsilon}}_{\text{A}} + \underbrace{1}_{\text{B}}\Big)\Delta\varepsilon + \lambda_B\Big(\underbrace{\frac{1}{n}\frac{\partial n}{\partial T}}_{\text{C}} + \underbrace{\frac{1}{\Lambda}\frac{\partial \Lambda}{\partial T}}_{\text{D}}\Big)\Delta T.$$

The contributing parts are due to various well known effects:

- **A** elasto-optic effect (change in optical properties of medium due to strain)
- **B**: elastic effect (relative change in periodicity due to strain)
- **C**: thermo-optic (change in optical properties of medium due to temperature)
- **D**: thermal expansion.

These composite factors of $\Delta\varepsilon$ and $\Delta T$ can be resumed respectively in the linear coefficient of strain $K_\varepsilon$, and in the coefficient of temperature $K_T$.

$$\Delta\lambda = K_\varepsilon\Delta\varepsilon + K_T\Delta T \tag{3}$$

An important point to be made is that this first order approximation allows us to decouple completely the two effects: once the rest point is fixed, the two coefficients are constant with respect to the two quantities, and there is not cross dependence.

## II. Experimental setup

In general, the measure of the peak wavelength of the reflected wave can follow two methods: in the first one a broadband CW source is injecting power in the fiber, and the reflected wave is analyzed by an optical spectrum analyzer, which, by means of a sweeping a monochromator over the wavelength range, measure the response for each wavelength. The second method is to employ a tunable laser as a source, set to sweep the wavelength periodically along the range, and a simple photodiode to measure the reflection at each wavelength.

In the current setup we are using an instrument which integrates the functionalities required by the second method, with a sweep frequency of 500 Hz, which is able to send the photodiode response data to a PC interface. In the sensor under test side, a couple of FBG sensors are connected in series, and then the fiber is left unconnected in the far side. Although the sensors are set to be in very close temperature condition, the first one is tied to the optical bench by means of a vise equipped with a transverse micrometer screw, set to impose strain on the sample, whereas the second is just leaning on the bench, as shown in Fig. 1.

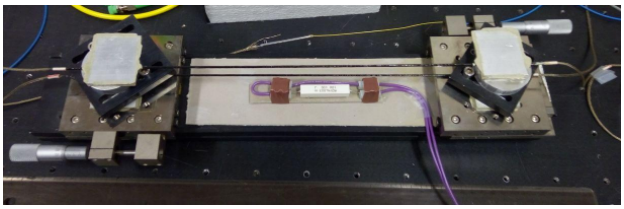

Fig. 1: Optical bench with the two sensors.

The sensors are chosen to have different rest-point Bragg wavelengths so they don't interact with each other, and we can measure separately the reflected peak wavelength without cross-interaction. Their $K$ coefficients are equal: this allows us to make interesting considerations on the simultaneous measuring of both strain and temperature. In fact, also the sensors lengths are equal, and are 31.5 cm.

The measurement campaigns are organized as follows:

**1)** The fibers are kept at the laboratory ambient temperature, and the wavelengths are measured for a set of strains, which are imposed by turning the micrometer screw from the rest condition to 500 µm, with steps of 50 µm. The measure along the set is repeated 3 times in order to check if the coefficient is subject to hysteresis. This first set is also used as a reference measurement for the second one.

| Parameter | Symbol | Set 1 | Set 2 | Set 3 | Average |
|---|---|---|---|---|---|
| Strain coefficient (sensitivity) [pm/με] | $K_\varepsilon$ | 11.313 | 11.316 | 11.347 | 11.326 |
| Squared residual [με$^2$] | $\rho^2_{min}$ | 37.561 | 1050.1 | 332.74 | 473.48 |
| Uncertainty of the responses [με] | $\sigma_r$ | 2.043 | 10.80 | 6.080 | 6.308 |
| Uncertainty of the sensor [pm] | $\sigma_s$ | 0.181 | 0.955 | 0.536 | 0.557 |
| Uncertainty of the parameter [pm/με] | $\epsilon_p$ | 0.007 | 0.035 | 0.020 | 0.020 |

TABLE I: Estimated parameters of the FBG sensor from the method of least squares at stationary temperature, for each set of measures; the statistical degree of freedom is $M - N = 9$ for every set.

**2)** The fibers are gradually heated up using a 10 Ω power resistor with dissipation of approximately 10 W. During the heating, the strain is imposed on the first fiber, with measures from the rest condition to 500 μm and steps of 100 μm. Even in this case the measure is repeated 5 times.

In Setup 2), the frequency shift of the first fiber is due to the compound effect of strain and temperature, whereas the second fiber is only subject to the temperature variation. The aim of the experiment is to obtain the sensor coefficient, and to isolate the same strain response exhibited by the first fiber in Setup 1) subtracting the response of the second fiber.

## III. Results

### A. Ambient Temperature

In this scenario, our setup was at STP (standard temperature and pressure) condition as we introduced in Section II. We collected both the back-scattered wavelengths from a reference fiber ($\lambda_R$) and a fiber under strain ($\lambda_S$). Naturally, the light reflected by the FBG behaves as described by (1), unlike the rest of the source spectrum that reaches the end of the fiber.

In Figure 2, we show the Bragg wavelength from the fiber which is under a gradually increasing strain. Since the reference fiber was not altered, we chose not to report the result from it because it is constant: $\lambda_R = 1533.1$ nm.

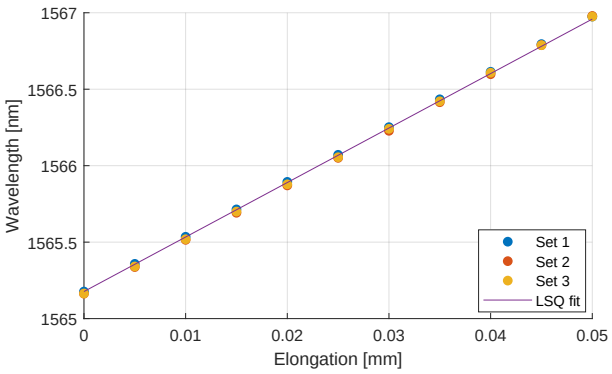


Fig. 2: Fiber under strain at room temperature.

Three sets of measures were collected; as it can be seen, all of them are virtually coincident. This means that there is no hysteresis, and gives proof of the linear relationship between elongation and wavelength, confirming the validity of a measure obtained with this sensor.

Assuming that the temperature is constant, the transfer function of the sensor shown in (3) simply becomes $\Delta\lambda = K_\varepsilon \Delta\varepsilon$. Also, under the same assumption, the vector of calibration parameters $\bar{p}$ is actually a scalar ($N = 1$) and corresponds to $p = K_\varepsilon$. Given the linearity of the sensor with respect to both the calibration parameter $K_\varepsilon$ and the stimulus $\Delta\varepsilon$, the estimation of the strain coefficient is straightforward, and can be done by using the method of least squares, which gives a closed-form expression to estimate the parameters and the uncertainties in the linear case. The transfer function is thus $\bar{r} = p\bar{s}$, where $\bar{s}$ and $\bar{r}$ are the vectors of the stimuli (i.e. the strain values) and the responses (i.e. the wavelength shifts), respectively.

After the estimation of $K_\varepsilon$, we were able to evaluate the uncertainty of the sensor and of the calibration parameter itself. First, the squared residual was estimated and used to compute the uncertainty on the responses. From $\sigma_r^2$ the uncertainty of the sensor was estimated as $\sigma_s \approx \sigma_r/|S|$, where $S$ is the sensitivity (which is constant and equal to $K_\varepsilon$). Similarly, it was possible to evaluate the covariance matrix of the model parameters, which was simply a scalar, $\epsilon_p$. The results of these calculations for all the sets of measures are summarized in Table I.

It can be noticed that, although the three sets looked almost coincident in the graph, there is a non-negligible deviation between the values. The increase in the uncertainty and the slight shift of the strain coefficient may be partially caused by a temperature drift between (and during) the measurements. Nonetheless, the calibration parameter is sufficiently consistent and still gives a very accurate result.

### B. Higher Temperature

In the second part of the experience, the fibers were gradually heated up. For both the fibers, we collected 5 samples of the same elongation (notation: $\lambda/1$ ... $\lambda/i$ ... $\lambda/5$). Unfortunately, the heat was not uniform for both of them, so the reference one (Fig. 3) shows weaker effects compared to the strained one (Fig. 4).

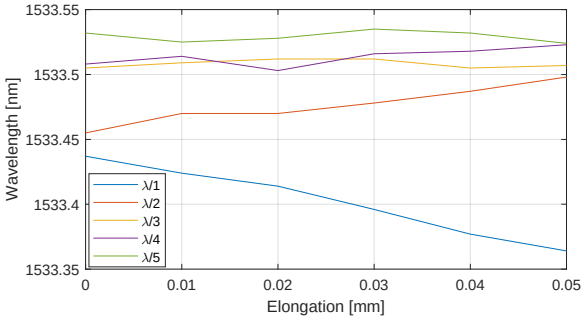


Fig. 3: Heated fiber (reference measurement).

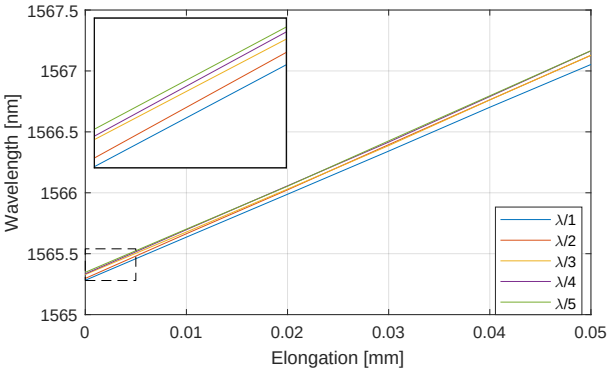


Fig. 4: Heated fiber under strain.

Moreover, the heat is not constant over time, since it slowly increases, thus increasing the Bragg wavelength, as well. This phenomenon is clearly visible in Figure 3, where the curve is shaped as a ladder due to the forward and backward movement of the cart, but can also be noticed in the magnification in Figure 4.

However, as explained in Section I, we used the data collected in the first part of the experiment to evaluate the thermal contribution $K_T \Delta T$ to the wavelength shift. Unfortunately, the temperature coefficient $K_T$ cannot be evaluated, given the lack of temperature measurements.

First, for each value of the heated reference fiber, we computed the difference (factor $K_T \Delta T$) between itself ($\lambda_R^{hot}$) and the corresponding measurement from its STP counterpart ($\lambda_R = 1533.1$ nm). Naturally, for further analysis we accounted for the fact that this difference is varying during time (increasing heat), but on average it is in the order of $0.02 \div 0.03\%$, so the average temperature contribution for the reference fiber is:

$$K_T \Delta T \approx 3 \cdot 10^{-4} \cdot \lambda_R^{hot} \approx 385 \text{ pm}.$$

Thanks to this linear scale factor (which is the same for both the fibers), we are able to translate the graph of the strain fiber to its true value, without the thermal wavelength shift (Fig. 5).

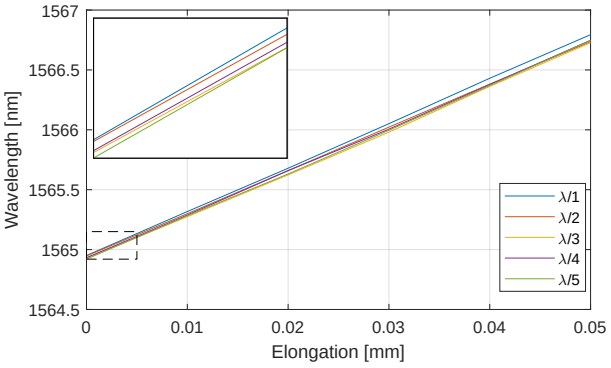


Fig. 5: Compensated heated fiber under strain.

Unfortunately, as already said, the heat was not uniform for both the fibers, hence instead of an ideal graph (Fig. 2) where all the lines are overlapped, some differences can still be seen. However, if we compare the magnified sections of Figure 4 and Figure 5, we can see that now the spacing between the lines is much narrower in the latter.

In order to compare the uncertainty between the STP fiber, the heated one and the corrected one, we evaluated the strain coefficient as a linear interpolation instead of using the least square method, which is unsuitable to account for thermal effects. For this reason the parameters of the first row in Table II are slightly different from the ones in Table I. The standard deviation is evaluated between the different $K_\epsilon$, each one obtained from the respective sets of measures.

| Fiber | $K_\epsilon$ [pm/μϵ] | STD [pm/μϵ] |
|---|---|---|
| STP fiber | 11.392 | 0.006 |
| Heated fiber | 11.428 | 0.159 |
| Corrected heated fiber | 11.404 | 0.146 |

TABLE II: Summary of the estimated strain coefficients in different scenarios.

As expected, after the correction, the corrected strain coefficient only slightly differs from the STP one, thus demonstrating the linearity and utility of FBGs as strain sensors.

Regarding the STD, it is superficial for the fiber at room temperature, since the lines in its graph are overlapped. As previously shown in Figure 5, the heated fiber is still not perfect after the correction but, as it can be seen (this time numerically) the error due to thermal expansion is reduced.

## IV. Conclusions

In this laboratory experience, we were able to observe the main working principles of FBGs, by measuring the wavelength shifts under different strain and temperature conditions.

The main takeaway from the first part of the experience was the effectiveness of the instrument, given that it was able to measure relatively small elongations with great accuracy. Actually, the estimated strain coefficient $K_\epsilon = 11.33$ pm/μϵ is approximately one order of magnitude larger than typical values found in this type of sensor (1.2 pm/μϵ), but our results appear to be consistent throughout the data. Assuming that the data processing is correct, the larger sensitivity may be a consequence of a different type of fiber.

On the other hand, in the second part, the temperature coefficient could not be estimated because the temperature variation was not controlled as the strain in the first part. Hence, the analysis was mainly focused on countering the intrinsic cross-sensitivity of FBGs, namely trying to separate the contribution to the wavelength shift caused by the strain from that caused by the temperature, which is still an unsolved problem for this type of sensor.

By subtracting a reference measure from a relaxed fiber, we were able to partially reduce the influence of temperature in the strain measurement, as the STD values show. Nonetheless, the temperature gradient in this part of the experiment was highly volatile, and large enough to make a difference between the two nearby fibers; as a matter of fact, the lines from different measurements are still distinguishable.